Appendix A
Intuitive method for vector calculus identities

Instead of providing the traditional "back of the book" list of vector calculus identities, an intuitive method[1] for deriving these identities will now be presented.

This method is based on combining the product rule of calculus with the vector algebra triple product and dot-cross product rules. The two vector rules will first be reviewed and then the method for combining these vector rules with the product rule of calculus will be presented.

Vector algebra triple product

There are two forms for the vector triple product, depending on the location of the parenthesis on the left-hand side, namely:

$$\mathbf{A}\times\left(\underbrace{\mathbf{B}}_{\text{middle}}\times\underbrace{\mathbf{C}}_{\text{outer}}\right)=\underbrace{\mathbf{B}}_{\text{middle}}\left(\underbrace{\mathbf{A}\cdot\mathbf{C}}_{\substack{\text{other two}\\ \text{dotted together}}}\right)-\underbrace{\mathbf{C}}_{\text{outer}}\left(\underbrace{\mathbf{A}\cdot\mathbf{B}}_{\substack{\text{other two}\\ \text{dotted together}}}\right) \tag{A.1}$$

$$\left(\underbrace{\mathbf{A}}_{\text{outer}}\times\underbrace{\mathbf{B}}_{\text{middle}}\right)\times\mathbf{C}=\underbrace{\mathbf{B}}_{\text{middle}}\left(\underbrace{\mathbf{A}\cdot\mathbf{C}}_{\substack{\text{other two}\\ \text{dotted together}}}\right)-\underbrace{\mathbf{A}}_{\text{outer}}\left(\underbrace{\mathbf{B}\cdot\mathbf{C}}_{\substack{\text{other two}\\ \text{dotted together}}}\right). \tag{A.2}$$

[1] This method was explained to the author by the late C. Oberman.

Both of these distinct forms can be remembered by the two-word mnemonic "middle-outer." The words middle and outer are defined with reference to the left-hand side of both equations; middle refers to the middle vector in the group of three and outer refers to the outer vector in the parentheses. The first term on the right-hand side is in the vector direction of the "middle" vector with the other two vectors dotted together; the second term on the right-hand side is in the direction of the "outer" vector with the other two vectors dotted together.

Dot-Cross product

Now consider the combination of dot and cross, $\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}$. Here the rules are that the dot and cross can be interchanged without changing the result and the order can be cyclically permuted without changing the result, but if the cyclic order is changed then the sign changes. Thus,

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=\mathbf{A}\times\mathbf{B}\cdot\,\mathbf{C} \quad \text{interchange dot and cross,} \tag{A.3}$$

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=\mathbf{B}\times\mathbf{C}\cdot\,\mathbf{A} \quad \text{permutation maintaining cyclic order,} \tag{A.4}$$

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=-\mathbf{A}\cdot\mathbf{C}\times\,\mathbf{B} \quad \text{permutation changing cyclic order.} \tag{A.5}$$

Derivation of the vector calculus identities

The basic idea is to replace one of the vectors by the ∇ operator and then rearrange terms and if necessary add terms. The criterion for these maneuvers is that, just like getting the right piece placed in a jigsaw puzzle, here all applicable rules of vector algebra and of calculus must be satisfied simultaneously. The simple example

$$\nabla\cdot(\psi\mathbf{A})=\psi\nabla\cdot\mathbf{A}+\mathbf{A}\cdot\nabla\psi \tag{A.6}$$

illustrates this principle of satisfying the vector algebra and the calculus rules simultaneously. Here the dot always goes between the ∇ and $\mathbf{A}$ in order to satisfy the rules of vector algebra and the ∇ operates once on $\mathbf{A}$ and once on ψ in order to satisfy the product rule $(ab)'=ab'+a'b$ of calculus.

A less trivial example is $\nabla\cdot(\mathbf{B}\times\mathbf{C})$. Here the ∇ must operate on both $\mathbf{B}$ and $\mathbf{C}$ according to the calculus product rule so, neglecting vector issues for now, the result must be of the form $\mathbf{B}\nabla\mathbf{C}+\mathbf{C}\nabla\mathbf{B}$. This basic product rule result is then adjusted to satisfy the vector dot-cross rules. In particular, the dots and crosses may be interchanged at will and the sign is plus if the cyclic order is $\nabla\mathbf{BC}$, $\mathbf{BC}\nabla$, or $\mathbf{C}\nabla\mathbf{B}$ and the sign is minus if the cyclic order is $\nabla\mathbf{CB}$, $\mathbf{CB}\nabla$, or $\mathbf{B}\nabla\mathbf{C}$. Since the ∇ is supposed to operate on only one vector at a time, we pick the arrangement where the ∇ is in the middle and either $\mathbf{B}$ or $\mathbf{C}$ is to its right so that each of $\mathbf{B}$ or $\mathbf{C}$ has a turn at being operated on by the ∇. The arrangements of interest are thus

$\mathbf{C}\cdot\nabla\times\mathbf{B}$ and $-\mathbf{B}\cdot\nabla\times\mathbf{C}$, where the minus sign has been inserted to account for the change in cyclic order. The result is

$$\nabla\cdot(\mathbf{B}\times C)=\mathbf{C}\cdot\nabla\times\mathbf{B}-\mathbf{B}\cdot\nabla\times\mathbf{C}, \tag{A.7}$$

which satisfies both the vector algebra dot-cross rule and the product rule of calculus.

Next consider the triple product $\mathbf{A}\times(\nabla\times\mathbf{C})$. Here the ∇ operates only on the $\mathbf{C}$ and the vector triple product generates two terms as indicated by Eq. (A.1). Expanding the triple product according to Eq. (A.1) while arranging an ordering of terms where the ∇ operates only on $\mathbf{C}$ gives

$$\mathbf{A}\times\left(\underbrace{\nabla}_{\text{middle}}\times\underbrace{\mathbf{C}}_{\text{outer}}\right)=\left(\underbrace{\nabla}_{\text{middle}}\underbrace{\mathbf{C}}_{\text{outer}}\right)\cdot\mathbf{A}-\left(\mathbf{A}\cdot\underbrace{\nabla}_{\text{middle}}\right)\underbrace{\mathbf{C}}_{\text{outer}}. \tag{A.8}$$

Thus, the first term on the right-hand side has its vector direction determined by ∇ with the other two terms dotted together, and the parenthesis indicates that the ∇ operates only on $\mathbf{C}$. The second term on the right-hand side has its vector direction determined by $\mathbf{C}$ with the other two terms dotted together and again the ∇ operates only on $\mathbf{C}$.

This can be rearranged as

$$(\nabla\mathbf{C})\cdot\mathbf{A}=\mathbf{A}\times(\nabla\times\mathbf{C})+\mathbf{A}\cdot\nabla\mathbf{C}. \tag{A.9}$$

Interchanging $\mathbf{A}$ and $\mathbf{C}$ gives

$$(\nabla\mathbf{A})\cdot\mathbf{C}=\mathbf{C}\times(\nabla\times\mathbf{A})+\mathbf{C}\cdot\nabla\mathbf{A}. \tag{A.10}$$

Adding these last two expressions gives

$$(\nabla\mathbf{C})\cdot\mathbf{A}+(\nabla\mathbf{A})\cdot\mathbf{C}=\mathbf{A}\times(\nabla\times\mathbf{C})+\mathbf{A}\cdot\nabla\mathbf{C}+\mathbf{C}\times(\nabla\times\mathbf{A})+\mathbf{C}\cdot\nabla\mathbf{A}. \tag{A.11}$$

However, using the same arguments about maintaining the dot and satisfying the product rule shows that

$$\begin{aligned}\nabla(\mathbf{A}\cdot\mathbf{C})&=\nabla(\mathbf{C}\cdot\mathbf{A})\\&=(\nabla\mathbf{A})\cdot\mathbf{C}+(\nabla\mathbf{C})\cdot\mathbf{A}.\end{aligned} \tag{A.12}$$

Here the ∇ operates once on the $\mathbf{A}$ and once on the $\mathbf{C}$ according to the product rule and the dot is always between the $\mathbf{A}$ and the $\mathbf{C}$. Combining Eqs. (A.11) and (A.12) gives the standard vector identity

$$\nabla(\mathbf{A}\cdot\mathbf{C})=\mathbf{A}\times(\nabla\times\mathbf{C})+\mathbf{A}\cdot\nabla\mathbf{C}+\mathbf{C}\times(\nabla\times\mathbf{A})+\mathbf{C}\cdot\nabla\mathbf{A}. \tag{A.13}$$

Finally, the triple product with two ∇'s can be expressed as

$$\nabla\times\left(\underbrace{\nabla}_{\text{middle}}\times\underbrace{\mathbf{A}}_{\text{outer}}\right)=\underbrace{\nabla}_{\text{middle}}\left(\underbrace{\nabla\cdot\mathbf{A}}_{\substack{\text{other two}\\ \text{dotted together}}}\right)-\underbrace{\nabla^2}_{\substack{\text{other two}\\ \text{dotted together}}}\underbrace{\mathbf{A}}_{\text{outer}} \tag{A.14}$$

or, without the labeling, as

$$\nabla\times\nabla\times\mathbf{A}=\nabla\nabla\cdot\mathbf{A}-\nabla^2\mathbf{A}. \tag{A.15}$$

The relationships $\nabla\cdot\nabla\times\mathbf{A}=0$ and $\nabla\times\nabla\psi=0$ can be proved by direct evaluation using Cartesian coordinates.

Summary of vector identities

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C})=\mathbf{B}\,(\mathbf{A}\cdot\mathbf{C})\,\mathbf{C}-(\mathbf{A}\cdot\mathbf{B})$$

$$(\mathbf{A}\times\mathbf{B})\times\mathbf{C}=\mathbf{B}\,(\mathbf{A}\cdot\mathbf{C})\,\mathbf{A}-(\mathbf{B}\cdot\mathbf{C})$$

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=\mathbf{A}\times\mathbf{B}\cdot\mathbf{C},\ \text{interchange dot and cross}$$

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=\mathbf{B}\times\mathbf{C}\cdot\mathbf{A},\ \text{cyclic permutation, cyclic order maintained}$$

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}=-\mathbf{A}\cdot\mathbf{C}\times\mathbf{B},\ \text{cyclic permutation, cyclic order changed}$$

$$\nabla\cdot(\psi\mathbf{A})=\psi\nabla\cdot\mathbf{A}+\mathbf{A}\cdot\nabla\psi$$

$$\nabla\cdot(\mathbf{A}\times\mathbf{B})=\mathbf{B}\cdot\nabla\times\mathbf{A}-\mathbf{A}\cdot\nabla\times\mathbf{B}$$

$$(\nabla\mathbf{B})\cdot\mathbf{A}=\mathbf{A}\times(\nabla\times\mathbf{B})+\mathbf{A}\cdot\nabla\mathbf{B}$$

$$\nabla\,(\mathbf{A}\cdot\mathbf{B})=\mathbf{A}\times(\nabla\times\mathbf{B})+\mathbf{A}\cdot\nabla\mathbf{B}+\mathbf{B}\times(\nabla\times\mathbf{A})+\mathbf{B}\cdot\nabla\mathbf{A}$$

$$\nabla\times\nabla\times\mathbf{A}=\nabla\,(\nabla\cdot\mathbf{A})-\nabla^2\mathbf{A}$$

$$\nabla\cdot\nabla\times\mathbf{A}=0$$

$$\nabla\times\nabla\psi=0$$